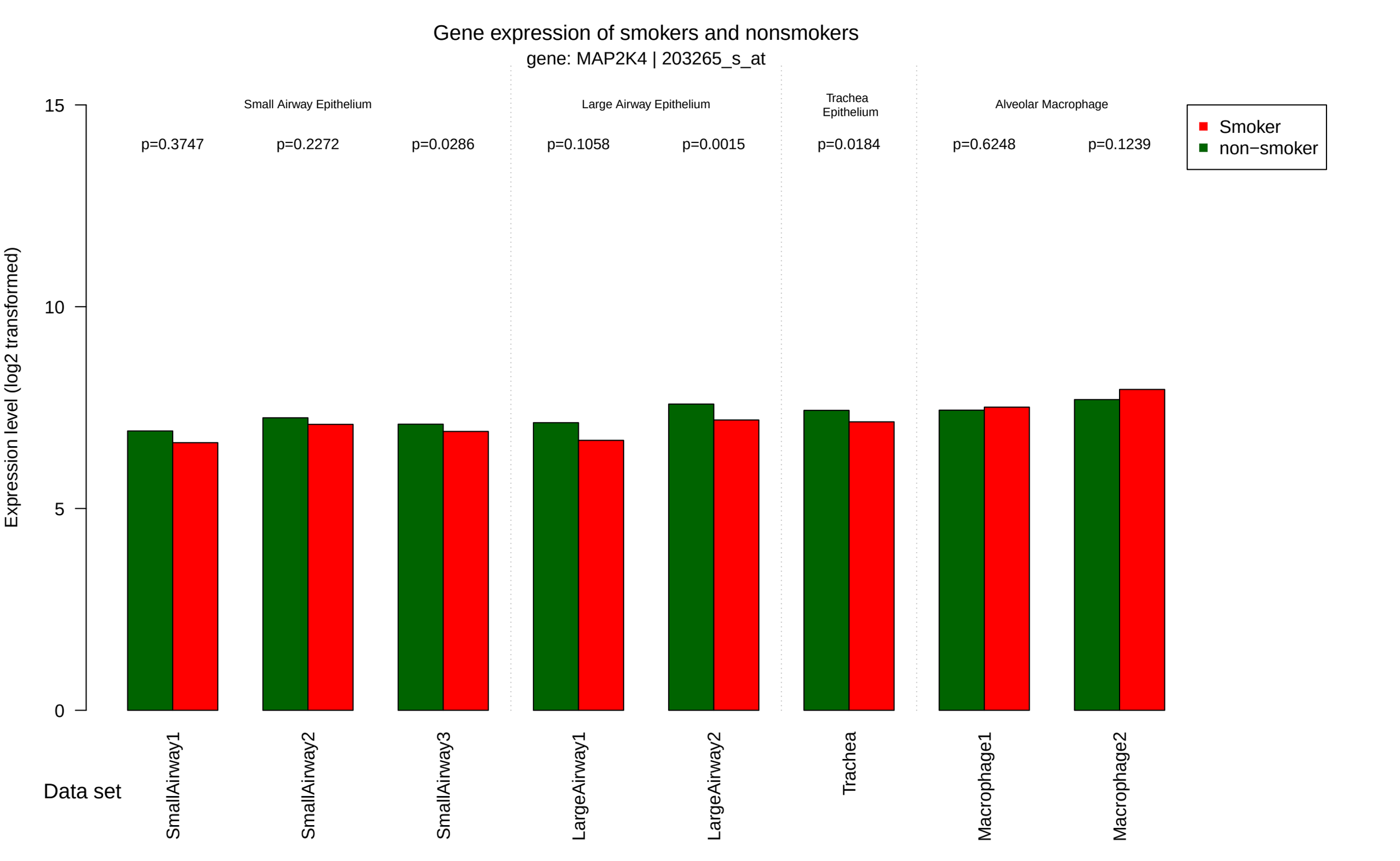
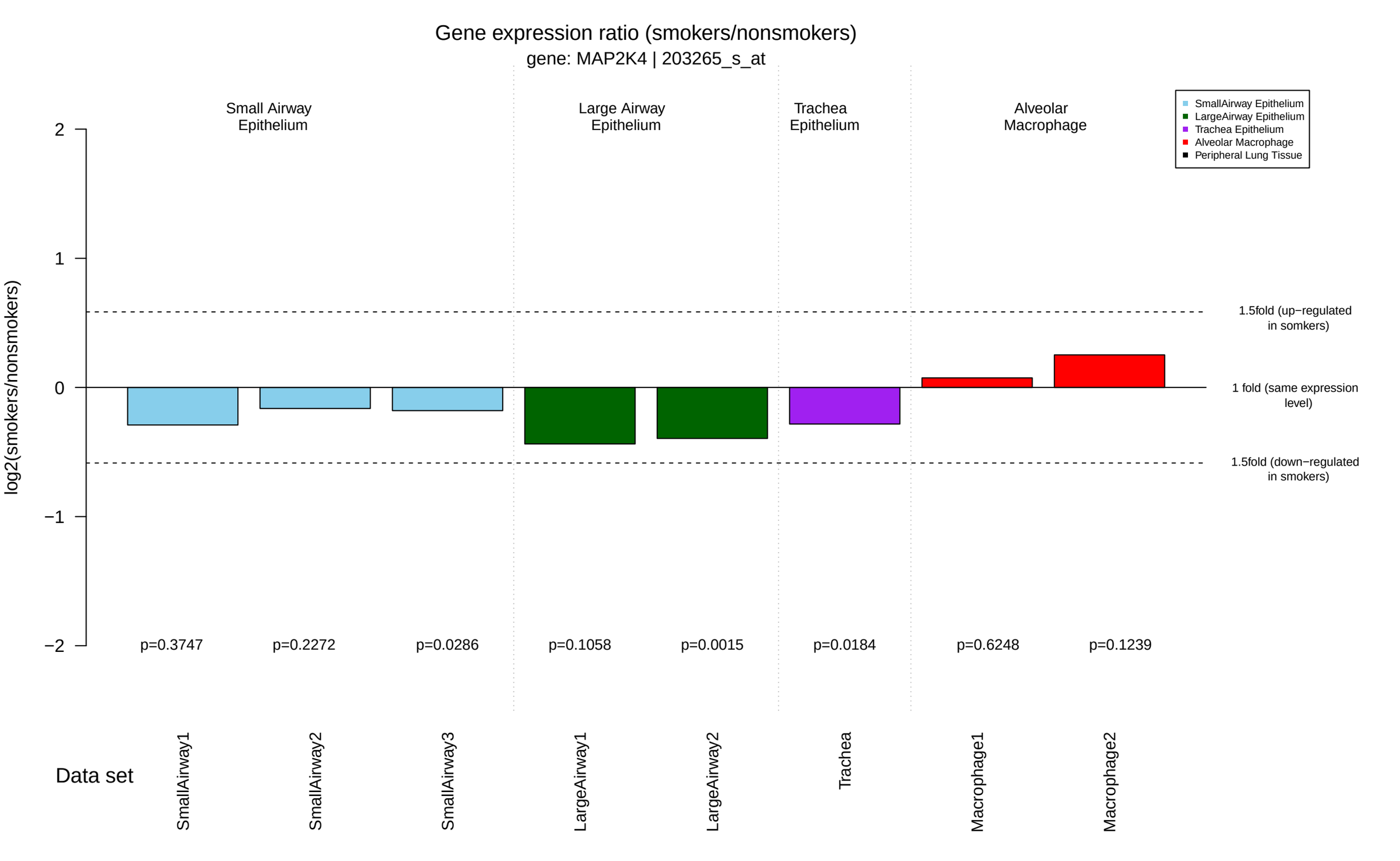
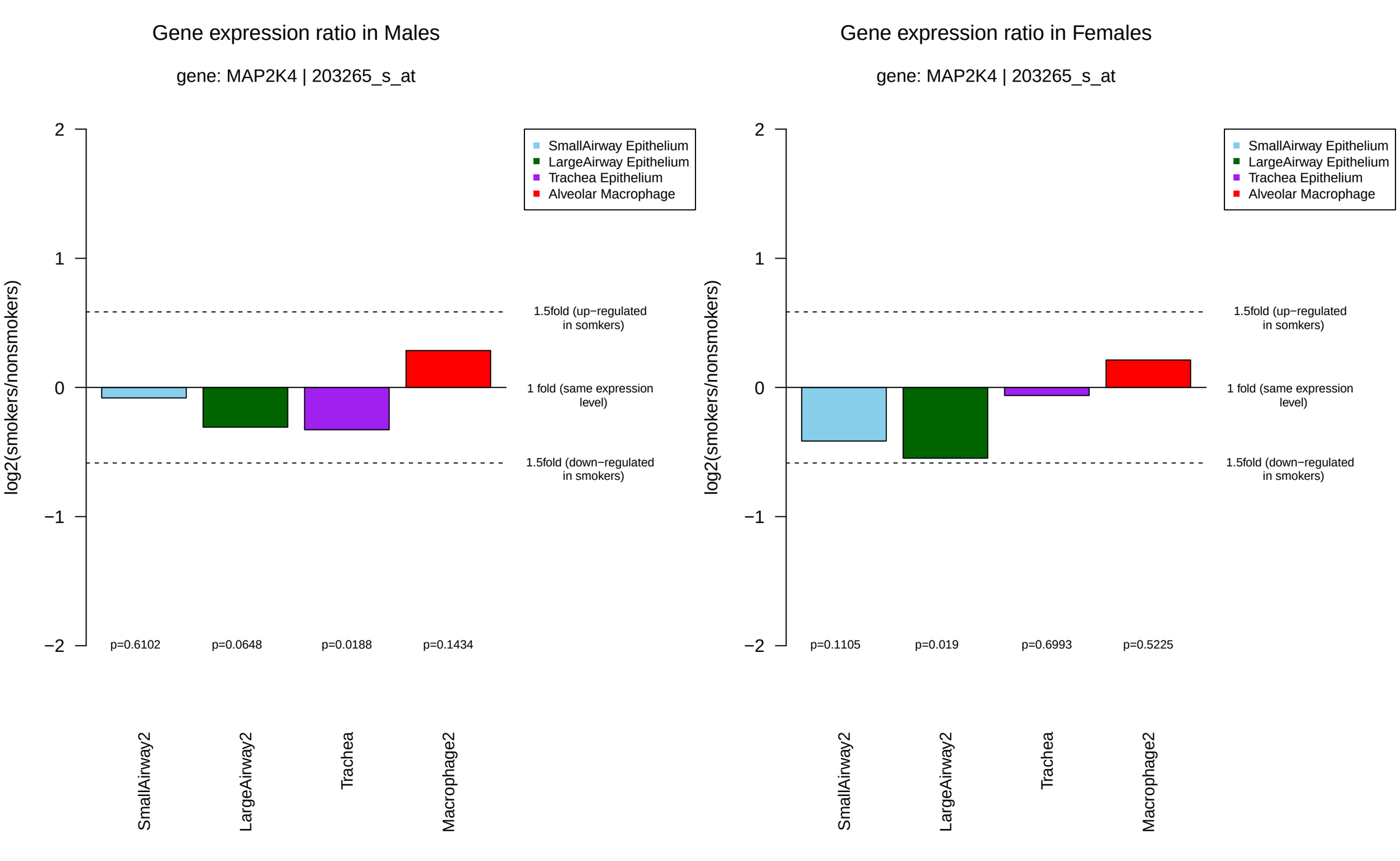
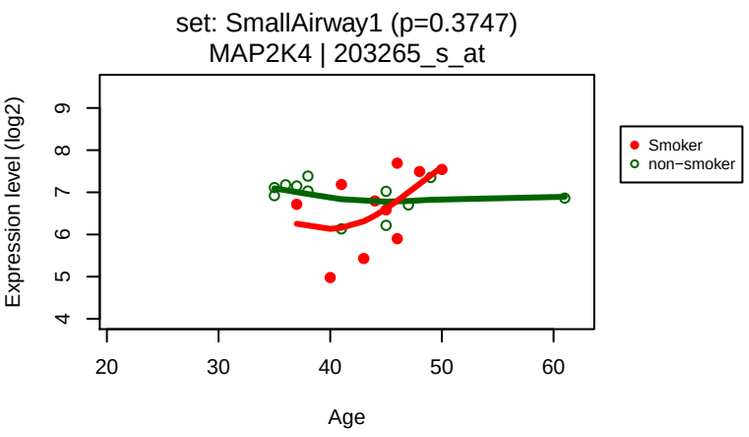
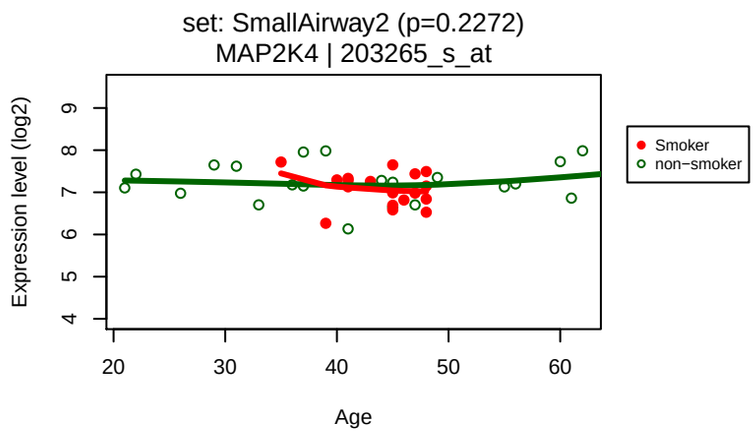
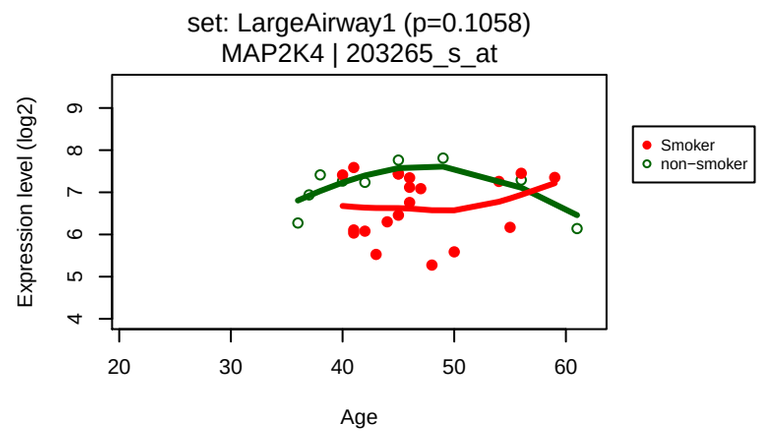

Gene expression of smokers and nonsmokers
gene: MAP2K4 | 203265_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3747
p=0.2272
p=0.0286
p=0.1058
p=0.0015
p=0.0184
p=0.6248
p=0.1239
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

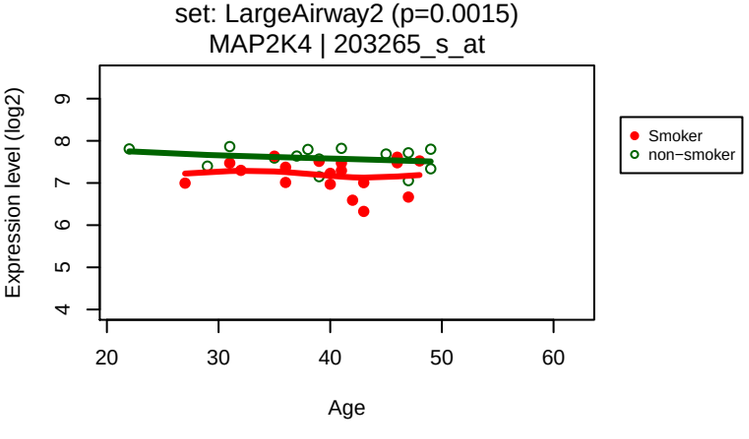

Gene expression ratio (smokers/nonsmokers)
gene: MAP2K4 | 203265_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3747
p=0.2272
p=0.0286
p=0.1058
p=0.0015
p=0.0184
p=0.6248
p=0.1239
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

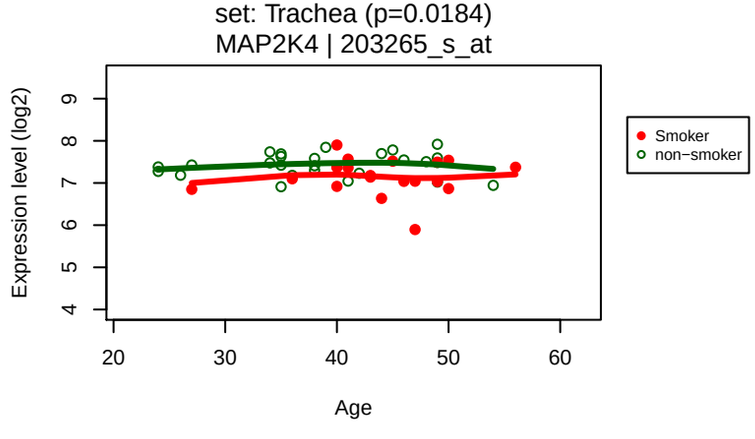

Gene expression ratio in Males
gene: MAP2K4 | 203265_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6102
p=0.0648
p=0.0188
p=0.1434
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MAP2K4 | 203265_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1105
p=0.019
p=0.6993
p=0.5225
SmallAirway2
LargeAirway2
Trachea
Macrophage2

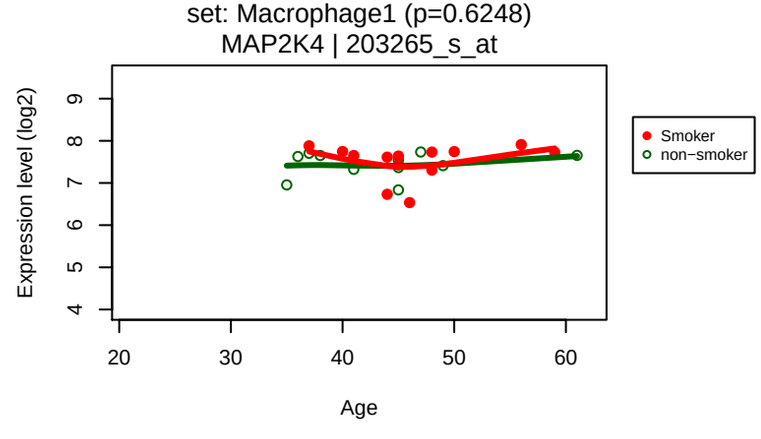

set: SmallAirway1 (p=0.3747)
MAP2K4 | 203265_s_at
Expression level (log2)
Age
Smoker
non−smoker

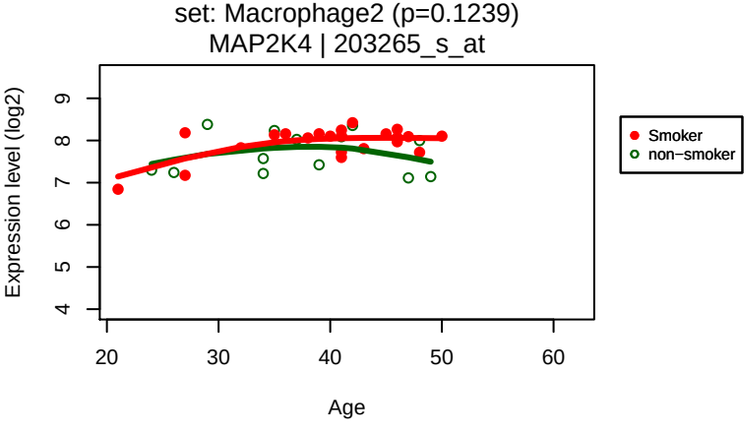

set: SmallAirway2 (p=0.2272)
MAP2K4 | 203265_s_at
Expression level (log2)
Age
Smoker
non−smoker

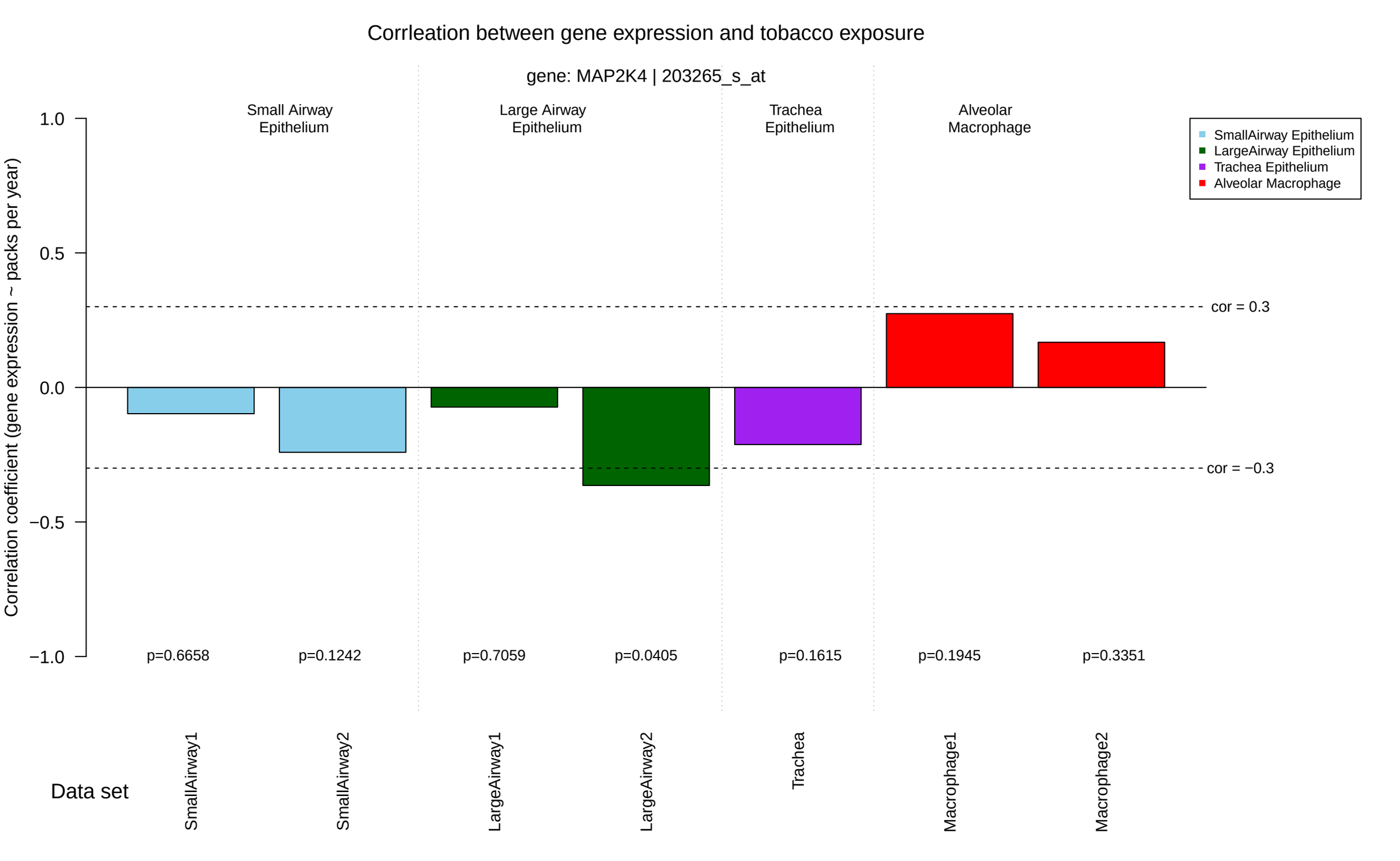

set: LargeAirway1 (p=0.1058)
MAP2K4 | 203265_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0015)
MAP2K4 | 203265_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0184)
MAP2K4 | 203265_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6248)
MAP2K4 | 203265_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1239)
MAP2K4 | 203265_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MAP2K4 | 203265_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6658
p=0.1242
p=0.7059
p=0.0405
p=0.1615
p=0.1945
p=0.3351
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2